

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

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Term: I

Roll No.: 1012411011

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Subject: Java

Assignment No.: 2

Title: Student Management System

Date: 04/08/2025

Java
Assignment 2

Name: Riddhee Kulkarni

PRN No: 1012411011

Aim: ₹

Implement a class with multiple constructors to demonstrate constructor overloading, specifically within the framework of a student management system.

Theory:

1. Constructor

A constructor in java is a special method that is used to initialize objects. It is called when an object of a class is created.

Syntax:

```
class Student  
{
```

```
    Student()  
{
```

```
        // Default constructor  
    }
```

```
}
```

```
class Student  
{
```

```
    Student (String name, int age)  
{
```

```
        name = name;
```

```
        age = age;
```

```
        // Parameterized constructor  
    }
```

```
}
```

2. Copy constructor

We can create a copy constructor by passing an object of the same class as a parameter.

Syntax:

```
Student (Student S)
{
    name = S.name;
    age = S.age;
}
```

3. Inner class

It is a class defined within another class.

Syntax:

```
class Outer
{
    class Inner
    {
        void fn()
        {
            //Statement
        }
    }
}
```

4. Enumeration

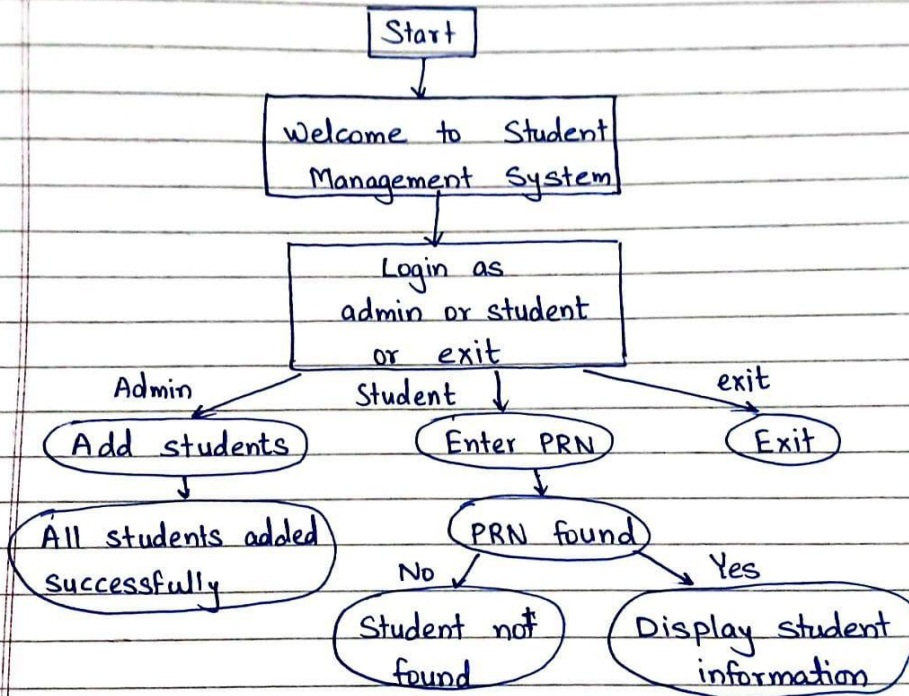
Enum is a special java type used to define a collection of constants.

eg. enum Days

```
{
    MON, TUES, WED, THURS, ...
}
```

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Flow chart:



Code:

```
import java.util.Scanner;

public class StudentManagement
{
    static Scanner sc = new Scanner(System.in);
    static StudentDatabase database = new StudentDatabase();
    static int totalStudents = 0;

    public static void main(String[] args)
    {
        String login;
        System.out.println("Welcome to Student Management System");

        do
        {
            System.out.print("\nLogin as (admin/student) or type 'exit' to quit: ");
            login = sc.nextLine();

            if (login.equals("admin"))
            {
                adminLogin();
            }
            else if (login.equals("student"))
            {
                studentLogin();
            }
            else if (!login.equals("exit"))
            {
                System.out.println("Invalid login type.");
            }
        } while (!login.equals("exit"));

        System.out.println("Exiting Student Management System. Goodbye!");
    }

    static class StudentDatabase {
        static Student[] students = new Student[100];
    }
}
```


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```
enum Year {
    FY, SY, TY;
}

public static void adminLogin()
{
    System.out.print("How many students do you want to add? ");
    int count = sc.nextInt();
    sc.nextLine();

    for (int i = 0; i < count; i++)
    {
        System.out.println("\nEnter details for student " + (i + 1) + ":");
        System.out.print("Name: ");
        String name = sc.nextLine();
        System.out.print("School: ");
        String school = sc.nextLine();
        System.out.print("Branch: ");
        String branch = sc.nextLine();
        System.out.print("Year (1 - FY, 2 - SY, 3 - TY): ");
        int yearNum = sc.nextInt();
        Year year = null;

        if (yearNum == 1) {
            year = Year.FY;
            StudentDatabase.students[totalStudents] = new Student(name, school, branch, year);
        } else if (yearNum == 2) {
            year = Year.SY;
            System.out.print("SGPA: ");
            double sgpa = sc.nextDouble();
            System.out.print("CGPA: ");
            double cgpa = sc.nextDouble();
            StudentDatabase.students[totalStudents] = new Student(name, school, branch, year, sgpa, cgpa);
        } else if (yearNum == 3) {
            year = Year.TY;
            System.out.print("SGPA: ");
            double sgpa = sc.nextDouble();
            System.out.print("CGPA: ");
```

```
            double cgpa = sc.nextDouble();
            StudentDatabase.students[totalStudents] = new Student(name, school, branch, year, sgpa, cgpa);
        } else {
            System.out.println("Invalid year. Skipping this student.");
            sc.nextLine();
            continue;
        }
        totalStudents++;
        sc.nextLine();
    }

    System.out.println("\nAll students added successfully.");
}

public static void studentLogin()
{
    System.out.print("Enter your PRN: ");
    int inputPRN = sc.nextInt();
    sc.nextLine();
    boolean found = false;

    for (int i = 0; i < totalStudents; i++)
    {
        if (StudentDatabase.students[i].prn == inputPRN)
        {
            StudentDatabase.students[i].display();
            found = true;
            break;
        }
    }

    if (!found)
    {
        System.out.println("Student with PRN " + inputPRN + " not found.");
    }
}
}
```

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```
class Student
{
    static int prnCounter = 1000;
    int prn;
    String name;
    String school;
    String branch;
    StudentManagement.Year year;
    double sgpa;
    double cgpa;

    Student(String name, String school, String branch, StudentManagement.Year year)
    {
        this.prn = prnCounter++;
        this.name = name;
        this.school = school;
        this.branch = branch;
        this.year = year;
    }

    Student(String name, String school, String branch, StudentManagement.Year year, double sgpa, double cgpa)
    {
        this.prn = prnCounter++;
        this.name = name;
        this.school = school;
        this.branch = branch;
        this.year = year;
        this.sgpa = sgpa;
        this.cgpa = cgpa;
    }

    void display() {
        System.out.println("\n--- Student Information ---");
        System.out.println("PRN: " + prn);
        System.out.println("Name: " + name);
        System.out.println("School: " + school);
        System.out.println("Branch: " + branch);
        System.out.println("Year: " + year);

        if (year == StudentManagement.Year.FY) {
            System.out.println("Branch is visible for FY students.");
        } else {
            System.out.println("SGPA: " + sgpa);
            System.out.println("CGPA: " + cgpa);
        }
    }
}
```

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Output:

```
Welcome to Student Management System

Login as (admin/student) or type 'exit' to quit: admin
How many students do you want to add? 2

Enter details for student 1:
Name: Riddhee
School: Engineering
Branch: CSE
Year (1 - FY, 2 - SY, 3 - TY): 2
SGPA: 9.1
CGPA: 9

Enter details for student 2:
Name: Riya
School: Business
Branch: Fintech
Year (1 - FY, 2 - SY, 3 - TY): 1

All students added successfully.

Login as (admin/student) or type 'exit' to quit: student
Enter your PRN: 1000

--- Student Information ---
PRN: 1000
Name: Riddhee
School: Engineering
Branch: CSE
Year: SY
SGPA: 9.1
CGPA: 9.0

Login as (admin/student) or type 'exit' to quit: exit
Exiting Student Management System. Goodbye!

...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion:

This program demonstrates the concept of constructor overloading, enum usage and java inner classes